CEYLON RAIL : AI-DRIVEN INTELLIGENT RAILWAY OPERATIONS SYSTEM FOR SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
Narasinghe N.K.B.N.N	IT22217240	Multi-Objective Train Scheduling Optimization
Didulantha C.S	IT22051202	Real time train tracking and delay prediction
Muthukudaarachchi V.U	IT22126610	Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification
D.M.N.T Senawirathna	IT22151506	[Component 4 Advance Spatio Temporal Demand Forecasting and Adaptive Seat Allocation Using AI]

R26-IT-127

Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) Degree in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

Oct 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name	Student ID	Signature	Date
Narasinghe N.K.B.N.N	IT22217240		
Didulantha C.S	IT22051202		
Muthukudaarachchi V.U	IT22126610		
D.M.N.T Senawirathna	IT22151506		

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: Mrs. Chathurangika Kahadawarachchi		
Co-supervisor:		

ABSTRACT

Railway transportation in Sri Lanka faces challenges related to fluctuating passenger demand, train delays, ticket fraud, capacity utilization, and scheduling. This research presents **CeylonRail Intelligence**, an AI-driven railway decision-support system comprising four components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, ticket fraud detection and passenger verification, and multi-objective train scheduling optimization.

The proposed architecture combines machine learning and optimization techniques to analyse railway data, identify operational patterns, detect suspicious transactions, and generate scheduling alternatives. The components are designed to support coordinated decision-making across railway operations.

Experimental results report a mean absolute percentage error (MAPE) of 7.2% for LSTM-based demand forecasting, delay prediction accuracy of 93.4%, and fraud detection accuracy of 94.6%. The scheduling component reduced average delay from 12.1 to 6.3 minutes in the reported experiments. The integrated system evaluation also reported a 99.1% integration success rate and an average API response time of 1.7 seconds. However, further testing with real-world railway data and broader validation is required before operational deployment. Overall, CeylonRail Intelligence demonstrates the potential of combining AI-based prediction, fraud detection, and optimization to support more efficient railway planning and decision-making in Sri Lanka.

Keywords: Intelligent Railway Systems, Artificial Intelligence, Passenger Demand Forecasting, Railway Fraud Detection, Train Scheduling Optimization.

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our research supervisor, Chathurangika Kahadawaarachchi, and our co-supervisor, Uditha Dharmakeerthi, for their valuable guidance, constructive feedback, continuous encouragement, and academic support throughout the development of the CeylonRail Intelligence research project. Their advice and recommendations contributed to refining the research direction, improving the proposed system, and strengthening the quality of this work.

We extend our appreciation to the lecturers, academic staff, and the Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic knowledge, learning environment, and resources that supported the completion of this research project.

We would also like to thank our fellow group members for their collaboration, cooperation, and shared commitment throughout the research, development, testing, and integration activities. The exchange of ideas and the coordination of the four individual research components were important in developing the proposed integrated railway intelligence system.

Finally, we express our heartfelt gratitude to our families and friends for their patience, understanding, encouragement, and continued support throughout our academic journey. Their motivation and support helped us overcome challenges and remain committed to completing this research project.

We are sincerely grateful to everyone who contributed to the successful completion of the CeylonRail Intelligence research project.

TABLE OF CONTENTS

ABSTRACT ii

ACKNOWLEDGEMENTS iii

TABLE OF CONTENTS iv

1 INTRODUCTION AND PROJECT CONTEXT 1

1.1 Background and Context 1

1.2 Stakeholders and User Needs 2

1.2.1 Primary Stakeholders 2

1.2.2 Secondary Stakeholders 2

1.2.3 Summary of Stakeholder Needs 3

1.3 Alignment with the Sustainable Development Goals 3

1.3.1 SDG 9: Industry, Innovation and Infrastructure 3

1.3.2 SDG 11: Sustainable Cities and Communities 4

1.3.3 SDG 12: Responsible Consumption and Production 4

1.3.4 Summary of SDG Alignment 4

1.4 Purpose and Scope of the Common Report 5

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 6

2.1 Overall Problem Definition 6

2.2 Root Causes and Overall Research Gap 6

2.2.1 Passenger Demand and Capacity Variability 6

2.2.2 Uncertainty in Train Movements and Delays 7

2.2.3 Suspicious Ticketing Activities 7

2.2.4 Scheduling and Resource Coordination 7

2.2.5 Fragmentation of Railway Intelligence 7

2.3 Subproblem and Component Mapping 8

2.3.1 Subproblem 1: Passenger Demand Forecasting and Adaptive Seat Allocation 8
2.3.2 Subproblem 2: Train Tracking and Delay Prediction 8
2.3.3 Subproblem 3: AI-Based Railway Fraud Detection and Passenger Verification 9
2.3.4 Subproblem 4: Multi-Objective Train Scheduling Optimization 9
2.3.5 Relationship Between the Four Subproblems 9
2.4 Overall Aim and Group Objectives 10
2.4.1 Overall Research Aim 10
2.4.2 Group Research Objectives 10
2.4.3 Objective-to-Component and Evaluation Mapping 11
2.5 Scope Assumptions and Constraints 12
2.5.1 Project Scope 12
2.5.2 Data Assumptions 12
2.5.3 Technical and Operational Constraints 12
2.5.4 Assumptions Concerning System Use 13
2.5.5 Exclusions and Boundaries 13
2.5.6 Summary of Scope and Constraints 13
3: OVERALL SOLUTION STRUCTURE 15
3.1 Integrated Solution Overview 15
3.2 High-Level Architecture 15
3.3 Component 1: Passenger Demand Forecasting and Adaptive Seat Allocation 17
3.4 Component 2: Real-Time Train Tracking and Delay Prediction 17
3.5 Component 3: AI-Based Railway Fraud Detection and Passenger Verification 18
3.6 Component 4: Multi-Objective Train Scheduling Optimization 18

3.7 End-to-End User and Data Flow 19
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 20
4.1 Integration Strategy 20
4.1.1 Integration Approach 20
4.1.2 Integration Sequence 20
4.1.3 Interface Ownership and Responsibilities 21
4.1.4 Integration Risks and Mitigation 21
4.2 Interfaces, Data Contracts and Shared Models 22
4.2.1 Interface Design 22
4.2.2 Data Contracts 22
4.2.3 Shared Identifiers and Data Consistency 23
4.2.4 Validation and Versioning 23
4.3 Shared Non-Functional Requirements 24
4.3.1 Performance and Responsiveness 24
4.3.2 Reliability and Availability 24
4.3.3 Security and Privacy 24
4.3.4 Usability and Accessibility 24
4.3.5 Maintainability and Scalability 25
4.3.6 Interoperability and Responsible Operation 25
4.4 Integrated Workflow and Exception Handling 25
4.4.1 Normal Integrated Workflow 25
4.4.2 Invalid or Incomplete Data 26
4.4.3 Component or Service Failure 26
4.4.4 Conflicting or Outdated Information 26
4.4.5 Fraud Alerts and Operational Safety 26
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 27

5.1 Implementation Overview 27
5.2 Development Environment and Technology Stack 27
5.2.1 Technology Selection 27
5.2.2 Frontend Implementation 28
5.2.3 Backend and Analytical Processing 28
5.2.4 Database and Data Storage 28
5.3 Implementation of the Four Research Components 29
5.3.1 Passenger Demand Forecasting and Adaptive Seat Allocation 29
5.3.2 Real-Time Train Tracking and Delay Prediction 29
5.3.3 AI-Based Railway Fraud Detection and Passenger Verification 30
5.3.4 Multi-Objective Train Scheduling Optimization 31
5.4 Integrated Application and Component Communication 31
5.4.1 Application Integration 31
5.4.2 Cross-Component Data Exchange 31
5.4.3 Handling Integration Errors 32
5.5 Project Execution and Coordination 32
5.5.1 Work Organization 32
5.5.2 Coordination and Version Control 33
5.5.3 Implementation Challenges and Responses 33
5.6 Deployment Considerations and Implementation Limitations 33
5.6.1 Deployment Considerations 33
5.6.2 Implementation Limitations 34
6 INTEGRATED TESTING, EVALUATION AND VALIDATION 35
6.1 Evaluation Framework and Success Criteria 35
6.2 Integration and End-to-End Testing 35
6.2.1 Testing Approach 35

6.2.2 Component and Interface Verification 36
6.2.3 End-to-End Behaviour 37
6.3 Overall Results 37
6.3.1 Passenger Demand Forecasting 37
6.3.2 Train Tracking and Delay Prediction 38
6.3.3 Fraud Detection and Passenger Verification 39
6.3.4 Multi-Objective Train Scheduling Optimization 40
6.3.5 Integrated System Performance 40
6.4 Stakeholder and User Validation 41
6.5 Achievement of Common Objectives 42
6.6 Integrated Limitations and Threats 43
6.7 Chapter Summary 44
7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION 45
7.1 Value Proposition and Beneficiaries 45
7.1.1 Primary Beneficiaries 45
7.1.2 Value Created by Integration 45
7.1.3 Social Value 46
7.2 Market Alternatives and Differentiation 46
7.2.1 Target Market 46
7.2.2 Existing Alternatives 46
7.2.3 Differentiation of the Integrated Solution 47
7.3 Business Model and Feasibility 47
7.3.1 Proposed Business Model 47
7.3.2 Technical Feasibility 48
7.3.3 Operational and Financial Feasibility 48

7.4 Adoption, Intellectual Property and Compliance 49
7.4.1 Adoption Barriers 49
7.4.2 Intellectual Property and Ownership 49
7.4.3 Data Protection, Privacy and Security 50
7.4.4 Responsible AI and Professional Responsibility 50
7.5 Sustainable Development Goal Contribution 51
7.5.1 SDG 9: Industry, Innovation and Infrastructure 51
7.5.2 SDG 11: Sustainable Cities and Communities 51
7.5.3 SDG 12: Responsible Consumption and Production 51
7.5.4 Potential Negative Effects and Sustainability Measures 51
7.5.5 Long-Term Sustainability 52
8 CONCLUSION AND FUTURE DIRECTIONS 53
8.1 Overall Conclusion 53
8.2 Principal Integrated Contribution 54
8.3 Future Directions 55
8.3.1 Reconcile Implementations and Evaluation Results 55
8.3.2 Improve Data Quality and Use Operational Data 55
8.3.3 Strengthen Demand Forecasting and Seat Allocation 55
8.3.4 Improve Delay Prediction and Tracking 56
8.3.5 Strengthen Fraud Detection and Responsible Verification 56
8.3.6 Expand Scheduling Evaluation 56
8.3.7 Complete Integration and Stakeholder Testing 56
8.3.8 Assess Deployment and Long-Term Sustainability 57
REFERENCES 58

LIST OF FIGURES

Figure 1 - High-Level Architecture of CeylonRail Intelligence 16

LIST OF TABLES

Table 1 - Stakeholder Needs and System Capabilities..... 3
Table 2 - Relationship Between CeylonRail Intelligence and Selected SDGs 4
Table 3 - Mapping of the Common Problem to the Research Components 9
Table 4 - Group Objectives and Evaluation Evidence 11
Table 5 - Scope Boundaries and Principal Constraints..... 14
Table 6 -Integration Stages and Expected Outputs 21
Table 7 - Proposed Shared Data Contracts 22
Table 8 - Technologies and Their Roles 27
Table 9 - Key Cross-Component Data Exchanges..... 32
Table 10 - Integrated Evaluation Framework 35
Table 11 - Integrated Test Matrix 36
Table 12 - Demand Forecasting Results 38
Table 13 - Delay Prediction Results 38
Table 14 - Integrated Fraud Detection Results 39
Table 15 - Scheduling Optimization Results 40
Table 16 - Integrated System Performance..... 40
Table 17 - Stakeholder Validation Status..... 41
Table 18 - Common Objectives 42
Table 19 - Potential Beneficiaries..... 45
Table 20 - Alternative Approaches and Potential Differentiation 46
Table 21 - Business Model Elements..... 48
Table 22 - Sustainability Measures 51

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
F1-score	F1 Measure (harmonic mean of precision and recall)
GPS	Global Positioning System
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NSGA-II	Non-dominated Sorting Genetic Algorithm II
RMSE	Root Mean Squared Error
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
R^2	Coefficient of Determination
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Railway transportation is an important part of public transport infrastructure, providing affordable mobility between cities and regional areas. Efficient railway operations require coordination of passenger demand, train capacity, service reliability, timetables, and operational resources. Artificial Intelligence (AI), Machine Learning (ML), predictive analytics, and optimization offer opportunities to improve railway management [1]–[4].

In Sri Lanka, passenger demand varies by route, travel period, weekday, season, and public holiday. These fluctuations can cause overcrowding and underutilized capacity, highlighting the need for accurate demand forecasting and effective seat allocation [5]–[10].

Train delays disrupt passenger journeys and operational coordination. Machine-learning and deep-learning techniques can analyse historical and operational data to predict delays and support planning [11], [12]. Ticket fraud is another challenge, as manual inspections and predefined rules may not identify every suspicious transaction. Machine-learning-based detection can prioritize unusual transactions for further review, while human verification remains essential [13]–[17].

Railway scheduling must balance timetables, capacity, congestion, infrastructure constraints, and service reliability. Multi-objective optimization can help evaluate competing operational requirements [18]–[20].

These challenges are interconnected: demand forecasts can inform capacity allocation and scheduling, while delay predictions and ticket-risk assessments can support operational decisions. In response, the group research project, **CeylonRail Intelligence: AI-Driven Intelligent Railway Operations System for Sri Lanka**, investigates four components:

1. Passenger demand forecasting and adaptive seat allocation.

2. Real-time train tracking and delay prediction.
3. AI-based railway fraud detection and passenger verification.
4. Multi-objective train scheduling optimization.

The proposed centralized architecture uses shared railway information, including routes, timetables, train capacities, demand patterns, ticketing policies, and operational constraints, to support coordinated analysis and decision-making. However, the reported experiments do not establish improved performance across Sri Lanka's railway network without further real-world validation.

1.2 Stakeholders and User Needs

The stakeholder needs described below are based on the project's identified challenges and intended functions, rather than confirmed findings from formal interviews or large-scale user studies.

1.2.1 Primary Stakeholders

- **Railway operations and planning personnel:** Need demand forecasts, capacity information, delay predictions, and scheduling alternatives to support operational planning.
- **Ticket inspection and revenue-protection personnel:** Need to identify suspicious transactions for further investigation. Automated risk scores support inspection but do not prove fraud.
- **Passengers:** Need reliable and accessible railway services. The system aims to support better service planning, although improvements in passenger satisfaction require validation.

1.2.2 Secondary Stakeholders

- **Railway management:** Needs information on demand, reliability, resource utilization, fraud risks, and scheduling performance.
- **System administrators and technical personnel:** Need maintainable interfaces, secure data handling, access controls, and performance monitoring.

- **Researchers and future developers:** Need a framework for evaluating and extending intelligent railway technologies, supported by reliable data and further testing.

1.2.3 Summary of Stakeholder Needs

Table 1 - Stakeholder Needs and System Capabilities

Stakeholder	Principal need	System capability
Railway operations personnel	Forecast demand and allocate capacity	Demand forecasting and seat allocation
Railway operations personnel	Monitor trains and predict delays	Train tracking and delay prediction
Inspection personnel	Identify suspicious transactions	Fraud detection and risk assessment
Railway planners	Evaluate scheduling alternatives	Multi-objective optimization
Passengers	Benefit from informed service planning	Combined component outputs
Technical personnel	Maintain a reliable platform	Shared architecture and data services
Researchers	Evaluate intelligent railway solutions	Integrated framework and experimental evidence

These capabilities address intended stakeholder needs but do not establish formal approval or measurable operational improvements.

1.3 Alignment with the Sustainable Development Goals

CeylonRail Intelligence has potential relevance to Sustainable Development Goals (SDGs) 9, 11, and 12. These represent intended contributions rather than demonstrated sustainability outcomes.

1.3.1 SDG 9: Industry, Innovation and Infrastructure

The project supports the goal conceptually by investigating AI-based railway monitoring, prediction, and scheduling. These capabilities may support data-driven operations, but their impact on infrastructure reliability requires real-world evaluation.

1.3.2 SDG 11: Sustainable Cities and Communities

Improved demand analysis, capacity allocation, and delay prediction could support more reliable public transport. However, passenger satisfaction, accessibility, road congestion, and emissions were not directly measured.

1.3.3 SDG 12: Responsible Consumption and Production

Demand forecasting and scheduling optimization aim to improve capacity and resource utilization. Actual savings depend on model accuracy, operational constraints, and implementation decisions.

1.3.4 Summary of SDG Alignment

Table 2 - Relationship Between CeylonRail Intelligence and Selected SDGs

SDG	Project relevance	Evidence needed
SDG 9: Industry, Innovation and Infrastructure	Intelligent railway monitoring and planning	Operational reliability and real-data validation
SDG 11: Sustainable Cities and Communities	Public transport planning	Passenger feedback and accessibility measures
SDG 12: Responsible Consumption and Production	Capacity and resource utilization	Resource consumption, utilization, and cost measurements

The project aligns conceptually with these goals. Demonstrating measurable contributions requires operational testing and sustainability monitoring.

1.4 Purpose and Scope of the Common Report

This report presents CeylonRail Intelligence as an integrated research and development project addressing interconnected railway challenges in Sri Lanka. It explains how four components—demand forecasting and seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and scheduling optimization—contribute to a common decision-support framework.

The report covers:

1. The common problem and research context.
2. Integrated architecture, shared data, and component relationships.
3. Implementation activities and project coordination.
4. Testing, evaluation results, and limitations.
5. Commercialization, sustainability, and responsible innovation.
6. Conclusions and future development.

Detailed datasets, algorithms, experimental procedures, and individual results are covered in the respective component research. The common report focuses on their relationship within the integrated system.

The findings represent experimental results under specific evaluation conditions, not proof of production-scale performance. Deployment would require real railway data, integration with existing railway systems, security and privacy assessments, and structured stakeholder evaluation.

Overall, the report examines the design, integration, and evaluated potential of CeylonRail Intelligence, emphasizing how four complementary intelligent capabilities could support more informed and efficient railway operations.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

Railway transportation requires coordinated management of passenger demand, train capacity, service reliability, ticketing, and scheduling. In Sri Lanka, fluctuating passenger demand, train delays, potentially fraudulent ticketing, and complex scheduling can affect passenger convenience, resource utilization, and operational efficiency.

Research has explored AI, machine learning, predictive analytics, and optimization for these challenges [3]–[20]. However, improving individual functions does not necessarily improve overall railway operations. Demand forecasts must inform capacity planning, delay predictions should support scheduling decisions, and fraud assessments require appropriate verification.

The common research problem is:

How can passenger demand and capacity management, train-delay prediction, railway ticket fraud detection, and train scheduling be coordinated through a data-driven decision-support framework suitable for Sri Lanka?

CeylonRail Intelligence addresses this problem through four components: demand forecasting and adaptive seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and multi-objective scheduling optimization. A centralized architecture is proposed to coordinate these capabilities through shared railway information and data services.

The project evaluates the feasibility and potential of this approach. Its findings do not establish nationwide operational effectiveness without further real-world validation.

2.2 Root Causes and Overall Research Gap

2.2.1 Passenger Demand and Capacity Variability

Passenger demand varies by route, travel period, passenger class, and season, potentially causing overcrowding or underutilized capacity. Forecasting methods can identify demand patterns [5]–[10], but predictions must also inform feasible seat allocations. The first component therefore combines demand forecasting with adaptive capacity allocation.

2.2.2 Uncertainty in Train Movements and Delays

Delays and variations in train movements complicate service monitoring and timetable coordination. Machine-learning and time-series methods can support delay prediction [10]–[12], but their usefulness depends on reliable data and integration with railway planning. The second component investigates train tracking and delay prediction.

2.2.3 Suspicious Ticketing Activities

Manual inspections and predefined rules may not identify every suspicious ticketing pattern. Machine-learning techniques can detect anomalies in transaction data [13]–[17]. However, unusual activity does not automatically indicate fraud. The third component investigates anomaly detection and risk assessment, with human verification retained for final decisions.

2.2.4 Scheduling and Resource Coordination

Railway scheduling must balance capacity, congestion, service reliability, infrastructure limitations, and passenger requirements. Multi-objective optimization can evaluate competing operational goals [18]–[20]. The fourth component investigates scheduling alternatives while considering relevant constraints.

2.2.5 Fragmentation of Railway Intelligence

The four challenges are interconnected. Demand forecasts can inform capacity allocation, delay predictions can support timetable evaluation, and fraud-risk assessments can assist revenue protection. The research gap addressed by CeylonRail

Intelligence is the coordination of these capabilities within a common railway decision-support framework [1]–[20].

The proposed architecture comprises user access, core intelligence, shared railway knowledge, data services, and database layers. Its purpose is to facilitate information exchange and coordinated analysis. The project's contribution is evaluated through available component-level and system-level evidence, rather than assumed operational benefits.

2.3 Subproblem and Component Mapping

2.3.1 Subproblem 1: Passenger Demand Forecasting and Adaptive Seat Allocation

This component addresses mismatches between passenger demand and available seating capacity. It investigates Random Forest and Long Short-Term Memory (LSTM) forecasting approaches alongside optimization-based seat allocation.

Inputs include historical passenger demand, route information, travel periods, and train capacity. Expected outputs are demand estimates and proposed seat allocations to support capacity planning and scheduling. Evaluation considers forecasting performance and allocation feasibility.

2.3.2 Subproblem 2: Train Tracking and Delay Prediction

This component addresses uncertainty in train movements and arrival times. It investigates Random Forest, Gradient Boosting, and LSTM approaches using operational features such as train movements, timetables, historical delays, station dwell times, and route congestion.

Expected outputs include predicted delays and estimated arrival times to support monitoring and timetable evaluation. Results based on historical or simulated GPS data require further validation before live operational use.

2.3.3 Subproblem 3: AI-Based Railway Fraud Detection and Passenger Verification

This component addresses the identification of suspicious ticketing activities. It investigates Isolation Forest, One-Class Support Vector Machine, and Autoencoder approaches, alongside risk scoring and rule-based verification.

Inputs may include transaction details, booking frequency, cancellations, route usage, and ticket-scanning behaviour. Outputs include anomaly assessments and cases requiring further review. Evaluation may use precision, recall, F1-score, ROC-AUC, and false-positive rate. Risk assessments support investigation rather than automatically confirm fraud.

2.3.4 Subproblem 4: Multi-Objective Train Scheduling Optimization

This component addresses competing scheduling objectives and operational constraints. It investigates the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to evaluate alternative schedules against measures such as average delay, capacity utilization, congestion, and resource utilization.

Demand forecasts and delay predictions may inform scheduling decisions. However, candidate schedules must be checked against railway rules, infrastructure limitations, and available operational data.

2.3.5 Relationship Between the Four Subproblems

Table 3 - Mapping of the Common Problem to the Research Components

Component	Main problem	Expected output	Integrated contribution

Demand forecasting and seat allocation	Demand uncertainty and capacity mismatches	Demand forecasts and proposed allocations	Capacity planning
Train tracking and delay prediction	Uncertain movements and delays	Delay predictions and arrival estimates	Monitoring and schedule evaluation
Fraud detection and verification	Suspicious ticketing activities	Risk assessments and cases for review	Ticket verification and revenue protection
Scheduling optimization	Competing objectives and constraints	Candidate schedules and performance comparisons	Coordinated scheduling

The intended relationships allow demand forecasts to inform capacity planning, delay predictions to support timetable evaluation, and fraud assessments to guide verification. Scheduling optimization evaluates alternatives using relevant operational information. Actual data exchanges must be supported by implementation and integration evidence.

2.4 Overall Aim and Group Objectives

2.4.1 Overall Research Aim

To design, develop, integrate, and evaluate an AI-driven railway decision-support system for Sri Lanka that combines passenger-demand forecasting and adaptive seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and multi-objective scheduling optimization.

2.4.2 Group Research Objectives

1. **Identify the common railway problem:** Examine challenges involving passenger demand, capacity, delays, ticket fraud, and scheduling.

2. **Develop four intelligent components:** Investigate and implement appropriate forecasting, prediction, detection, and optimization techniques.
3. **Establish a common architecture:** Design shared railway information services, databases, and component interfaces.
4. **Support coordinated decision-making:** Investigate how component outputs can inform related railway decisions.
5. **Evaluate system performance:** Assess model metrics, scheduling indicators, integration results, and technical performance.
6. **Identify deployment requirements:** Examine data limitations, operational constraints, security, privacy, scalability, and validation needs.

2.4.3 Objective-to-Component and Evaluation Mapping

Table 4 - Group Objectives and Evaluation Evidence

Objective	Relevant activities	Evaluation evidence
1. Define the common problem	Literature review and requirements analysis	Problem analysis and documented requirements
2. Develop components	Four research components	Implementations and experimental results
3. Establish architecture	Shared services and interfaces	Architecture documentation and integration tests
4. Support coordination	Component interactions	Data flows and workflow tests
5. Evaluate performance	Component and system evaluation	Model metrics, scheduling results, and performance measurements
6. Identify deployment needs	System-wide assessment	Limitations, security considerations, and future work

Achievement of each objective should be assessed against available evidence and classified as fully, partly, or not achieved. Component implementation alone does not prove operational effectiveness.

2.5 Scope Assumptions and Constraints

2.5.1 Project Scope

The research covers four capabilities:

- Passenger-demand forecasting and adaptive seat allocation.
- Train tracking and delay prediction.
- Ticket fraud detection and passenger verification.
- Multi-objective train scheduling optimization.

It also considers a shared architecture, component coordination, experimental evaluation, and future deployment requirements. The project represents a research prototype and decision-support framework, not a confirmed production deployment within Sri Lanka Railways.

2.5.2 Data Assumptions

The research assumes that available datasets contain relevant passenger, ticketing, timetable, movement, and scheduling information. Model performance depends on data completeness, accuracy, consistency, and representativeness.

Historical or simulated data may be used where live operational information is unavailable. Results must therefore be interpreted within the relevant experimental conditions. Cross-component analysis also requires compatible data formats and identifiers.

2.5.3 Technical and Operational Constraints

- **Data quality:** Missing, inaccurate, or unrepresentative records may affect model performance.
- **Real-time access:** Historical or simulated movement data do not establish live tracking reliability.

- **Integration:** Component coordination depends on compatible interfaces and data formats.
- **Model uncertainty:** Predictions may be inaccurate, anomaly detection may produce false positives, and optimized schedules require feasibility checks.
- **Operational restrictions:** Train capacities, safety requirements, railway rules, and infrastructure limitations must be considered.
- **Security and privacy:** Passenger and ticketing data require appropriate access controls and secure handling.
- **Stakeholder validation:** Formal, large-scale validation with railway personnel and passengers has not been established by the available research evidence.

2.5.4 Assumptions Concerning System Use

The system is intended to support, not replace, railway personnel. Predictions are estimates, fraud-risk scores prioritize further investigation, and proposed schedules require verification before implementation. Reliable deployment also requires suitable interfaces, failure handling, security, and compatibility with existing railway systems.

2.5.5 Exclusions and Boundaries

The current evidence does not establish:

1. Nationwide production deployment or operational certification.
2. Guaranteed elimination of delays, overcrowding, fraud, or scheduling inefficiencies.
3. Verified improvements in passenger satisfaction, staff productivity, or revenue.
4. Measured reductions in energy consumption or transport emissions.
5. Fully validated integration with all existing railway systems.
6. Automatic enforcement of fraud decisions or scheduling changes without authorization.

2.5.6 Summary of Scope and Constraints

Table 5 - Scope Boundaries and Principal Constraints

Area	Research coverage	Main constraint
Demand and capacity	Forecasting and seat allocation	Data quality and capacity constraints
Train delays	Movement analysis and prediction	Reliable operational data required
Fraud detection	Anomaly detection and risk assessment	Human review and false-positive management
Scheduling	Multi-objective optimization	Operational feasibility
Integration	Common architecture and coordination	Interface and data compatibility
Evaluation	Experimental and technical testing	Limited real-world validation
Deployment	Prototype and future requirements	Security and reliability assessments
Sustainability	Potential SDG alignment	Direct impact measurements required

In summary, CeylonRail Intelligence investigates how four complementary AI-driven capabilities can be coordinated within a railway decision-support framework. The defined scope and constraints clarify the current research contribution and the additional validation required before operational adoption.

3: OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

CeylonRail Intelligence is an AI-driven railway decision-support system proposed to address passenger demand fluctuations, seat utilization, train delays, ticket fraud, and scheduling challenges in Sri Lanka. It combines four components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and multi-objective train scheduling optimization.

The system aims to transform railway data into forecasts, risk indicators, and scheduling recommendations through a common platform. For example, demand forecasts can support capacity planning, while delay estimates can inform schedule evaluation. Ticket verification and fraud-risk assessment can help identify transactions requiring further review.

The platform is intended to support railway administrators, operational staff, ticket inspectors, and passengers through relevant information and services. Its outputs support human decision-making rather than replace railway authorities. The architecture and component implementations are described in the following sections and individual theses [3], [4].

3.2 High-Level Architecture

The proposed architecture consists of five layers: User and Platform Access, Core Intelligence, Shared Railway Knowledge, Data Services, and Centralized Database.

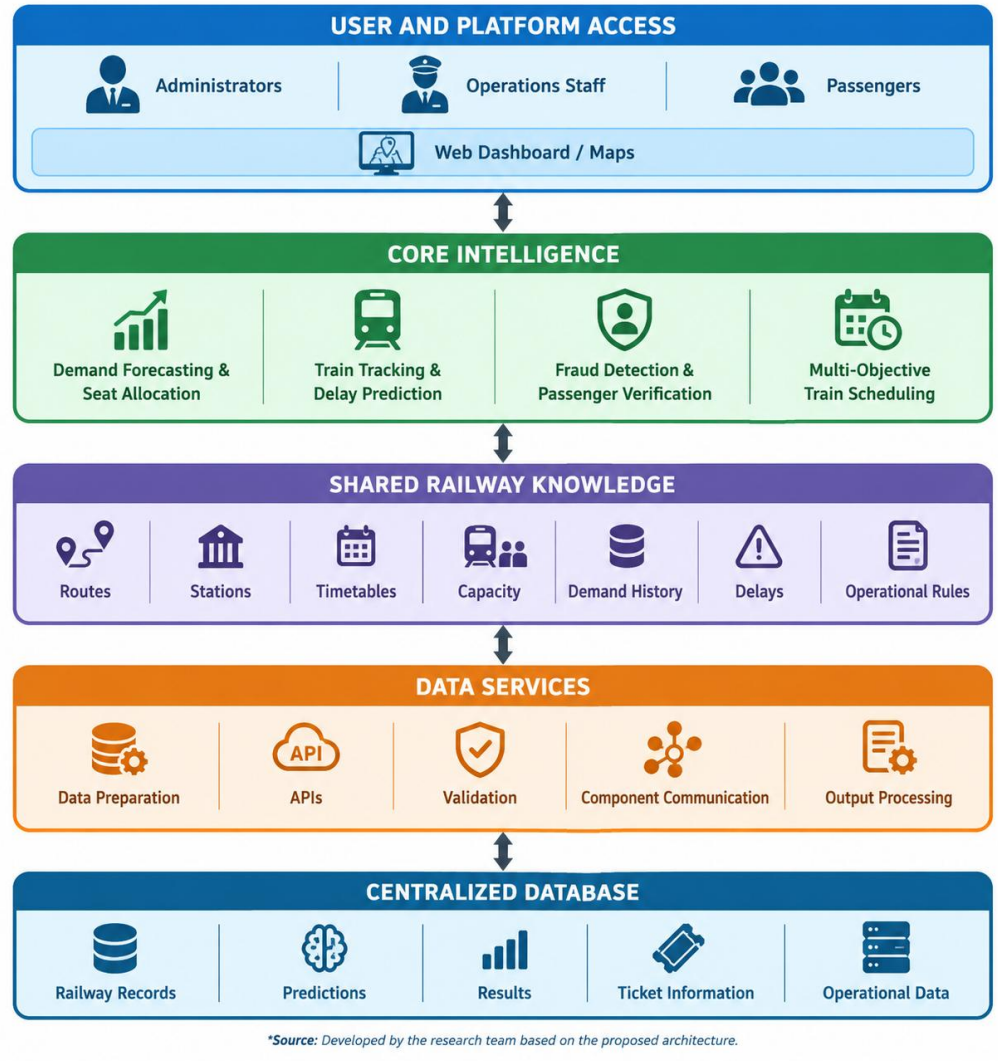


Figure 1 - High-Level Architecture of CeylonRail Intelligence

Figure 1 illustrates how users access system functions through the platform. The Core Intelligence Layer contains the four components, while the Shared Railway Knowledge and Data Services layers support common information, validation, communication, and processing. The Centralized Database provides storage for relevant railway records and analytical outputs.

This architecture aims to encourage consistent use of shared information. However, it represents the proposed logical design; live data availability and complete component integration require implementation evidence and testing.

3.3 Component 1: Passenger Demand Forecasting and Adaptive Seat Allocation

Component owner: Senawirathna D.M.N.T.

This component addresses changing passenger demand across routes, travel periods, passenger classes, and seasons. It uses historical ticket sales, passenger records, calendar features, route information, and available capacity to forecast demand and support seat allocation.

The individual research evaluates Random Forest and XGBoost, with LSTM investigated separately. Random Forest is identified as the preferred forecasting model in the individual thesis. The model used in any integrated workflow should remain consistent with the implementation being evaluated.

The outputs include demand estimates and proposed seat allocations that can support capacity planning and provide demand-related information to the scheduling component.

3.4 Component 2: Real-Time Train Tracking and Delay Prediction

Component owner: Didulantha C.S.

This component aims to improve train-status monitoring and delay estimation. Its inputs include train identifiers, timestamps, scheduled arrival and departure times, route information, movement data, and historical delays where available.

The processing workflow includes data validation, status updates, location-based presentation, and machine-learning-based delay prediction. Although the individual research investigates LSTM-based prediction, its classification experiment assigned all 1,065 test observations to the Moderate Delay category. This indicates a limitation in distinguishing delay classes and does not establish accurate live prediction.

Intended outputs include train status, mapped movement information, estimated arrival times, delay estimates, and alerts. Reliable real-time operation depends on timely and accurate railway data.

3.5 Component 3: AI-Based Railway Fraud Detection and Passenger Verification

Component owner: Muthukudaarachchi V.U.

This component aims to identify suspicious ticketing activity and support ticket and passenger verification. It uses available ticket information, QR validation records, transaction characteristics, travel history, route information, and railway rules.

The individual research investigates Isolation Forest, One-Class SVM, and Autoencoder methods alongside risk-based assessment. Intended outputs include ticket-validation results, suspicious-activity indicators, risk categories, and cases requiring further inspection.

Risk scores are decision-support signals, not conclusive proof of fraud. Human review, access controls, and passenger-data protection remain essential.

3.6 Component 4: Multi-Objective Train Scheduling Optimization

Component owner: Narasinghe N.K.B.N.N.

This component addresses scheduling trade-offs involving delays, capacity, congestion, and resource utilization. It considers timetable information, routes, stations, train capacity, predicted demand, operational constraints, and available delay information.

The component uses the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to explore alternative schedules across multiple objectives. Its outputs are intended to help decision-makers compare candidate schedules and understand trade-offs. Final decisions must comply with railway safety requirements and authorized operating procedures.

Where validated interfaces are available, the optimizer can use demand forecasts and delay estimates to support more coordinated planning.

3.7 End-to-End User and Data Flow

The proposed workflow connects railway data, component processing, and decision-support outputs through the shared platform.

1. **Data collection and preparation:** Obtain available railway, timetable, demand, movement, and ticketing records; validate and prepare the data.
2. **Component processing:** Send relevant information to the forecasting, tracking, fraud-detection, and scheduling components.
3. **Information exchange:** Share validated predictions, risk indicators, and operational summaries where interfaces have been implemented.
4. **Decision support:** Present demand forecasts, train status, verification alerts, and schedule alternatives through the platform.
5. **Human review:** Enable authorized personnel to assess outputs and make decisions under railway rules.
6. **Monitoring and improvement:** Retain appropriate outputs and evaluation records for future assessment, subject to data-quality and privacy requirements.

Demand forecasts can inform seat allocation, while train-status and delay information can support schedule evaluation. Ticket-verification results can help prioritize inspections, and scheduling optimization can compare alternatives using available operational inputs. Each component can also produce independent outputs without relying on every other component.

The practical effectiveness of this workflow depends on data availability, consistent identifiers, compatible interfaces, and verified information exchange. The following chapter discusses the integration strategy, shared data contracts, and coordination of the four components in greater detail.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

4.1.1 Integration Approach

CeylonRail Intelligence combines four research components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, AI-based fraud detection and passenger verification, and multi-objective train scheduling optimization.

The integration strategy preserves each component's analytical independence while enabling information exchange through shared railway data and defined interfaces. Demand forecasts can inform capacity planning, delay predictions can support scheduling, and fraud-risk assessments can assist authorized verification.

The strategy follows six principles:

- **Modularity:** Components maintain their individual processing logic.
- **Shared information:** Train, route, station, and timetable data follow consistent definitions.
- **Defined interfaces:** Components exchange information through agreed formats.
- **Loose coupling:** Components communicate through interfaces rather than internal implementations.
- **Data consistency:** Shared identifiers and validation rules reduce inconsistencies.
- **Human oversight:** Predictions and recommendations support authorized decisions.

4.1.2 Integration Sequence

Table 6 -Integration Stages and Expected Outputs

Stage	Main activity	Expected output
1. Requirements	Identify shared data and dependencies	Integration requirements
2. Interface design	Define formats and validation rules	Interface specifications
3. Component preparation	Standardize inputs and outputs	Testable components
4. Service integration	Connect components and shared services	Connected interfaces
5. Verification	Test data exchange and workflows	Integration evidence
6. Refinement	Resolve errors and inconsistencies	Improved compatibility

These stages describe the integration approach. Completed, partial, and planned activities must be distinguished using implementation evidence.

4.1.3 Interface Ownership and Responsibilities

Component owners maintain their processing logic, input validation, outputs, and documentation. The integration coordinator manages shared identifiers, interface conventions, and cross-component testing.

Demand forecasts should identify the relevant service and forecast period, while delay predictions should specify train identifiers and estimated arrival times. Fraud assessments and scheduling outputs should follow similarly defined formats.

Where components use different frameworks or databases, APIs or adapters may be needed. The final report should document the interfaces actually implemented.

4.1.4 Integration Risks and Mitigation

Potential risks include incompatible formats, missing records, inconsistent identifiers, and service failures. Mitigation measures include interface specifications, validation, version control, component testing, and regression testing.

Offline model performance must also be distinguished from integrated service performance because deployment may introduce additional data-processing and configuration requirements.

4.2 Interfaces, Data Contracts and Shared Models

4.2.1 Interface Design

Application programming interfaces (APIs) connect the user interface, data services, analytical components, and storage.

The proposed integrated technology stack includes React.js, Tailwind CSS, Leaflet Maps, Node.js/Express.js, Python, TensorFlow, Scikit-Learn, NSGA-II, and MongoDB. However, individual implementations differ; for example, the fraud-detection thesis describes React, Flask, Python, and MySQL. The report must distinguish the proposed architecture from the technologies actually implemented.

4.2.2 Data Contracts

A data contract defines exchanged fields, data types, identifiers, units, validation rules, and error responses.

Table 7 - Proposed Shared Data Contracts

Information	Important fields	Purpose
Demand forecasts	Service ID, route, forecast period, predicted demand	Communicate expected demand
Seat allocation	Service ID, capacity, allocated seats, utilization	Present allocation results

Train tracking	Train ID, station/location, timestamp, estimated arrival, predicted delay	Communicate train status
Fraud assessment	Transaction ID, validation result, risk category, assessment time	Support transaction review
Scheduling	Schedule ID, train/route IDs, constraints, objective values	Compare candidate schedules
Railway reference data	Train, route, station IDs, timetable, capacity	Maintain consistent information

Note: These contracts describe proposed requirements and do not confirm that every field is implemented.

Missing values must be distinguished from valid zero values. For example, a zero-minute delay is different from an unavailable prediction.

4.2.3 Shared Identifiers and Data Consistency

Consistent train, route, station, and service identifiers are essential for linking information across components. Data services should standardize date formats, timestamps, units, and naming conventions.

Observed, scheduled, and predicted times must remain distinguishable. Predictions should also identify the relevant service and period to ensure correct interpretation.

4.2.4 Validation and Versioning

Incoming data should be checked for required identifiers, valid formats, acceptable numerical ranges, and consistency. Interface changes should be version-controlled, communicated to dependent components, and tested before release.

Errors should identify invalid requests or unavailable information without exposing credentials or unnecessary passenger data.

4.3 Shared Non-Functional Requirements

4.3.1 Performance and Responsiveness

The platform should provide timely access to operational information. Performance depends on data volume, analytical complexity, database access, and service communication. Computationally intensive tasks, such as scheduling optimization, may require separate processing and status updates.

The integrated research paper reports an average API response time of 1.7 seconds and dashboard loading time of 1.4 seconds. These results apply to the reported test conditions and do not guarantee equivalent performance for every operation.

4.3.2 Reliability and Availability

The system should handle invalid inputs, service failures, and temporary network or database problems without displaying misleading results. Where possible, unaffected functions should remain available when a component fails.

Recovery procedures should address interrupted processing and restoration of persistent data.

4.3.3 Security and Privacy

Access should be restricted according to user roles and legitimate operational needs. Security measures include authentication, authorization, protected communications, secure configuration, and controlled database access.

Passenger information should be limited to necessary fields. Fraud-risk scores must not be treated as conclusive proof, and access to sensitive records should be restricted. Logging, retention, backup, and deletion practices should follow institutional and applicable data-protection requirements.

4.3.4 Usability and Accessibility

The interface should use consistent labels, units, status indicators, and clear validation messages. Important alerts should not rely on colour alone.

Observed information, predictions, and recommendations must be clearly distinguished to prevent estimated arrival times or fraud-risk assessments from being interpreted as confirmed facts.

4.3.5 Maintainability and Scalability

Clear service boundaries, documented interfaces, version-controlled configurations, and separation of data preparation from analytical processing support maintainability.

Scalability concerns handling additional records, trains, services, and users. It must be demonstrated through testing rather than assumed from the technology stack.

4.3.6 Interoperability and Responsible Operation

Different frameworks and databases require compatible interfaces and consistent data definitions. Human oversight must remain in place for consequential decisions. Scheduling recommendations require operational checks, while fraud alerts should support investigation rather than automatic punitive action.

4.4 Integrated Workflow and Exception Handling

4.4.1 Normal Integrated Workflow

The workflow begins when a user requests information or new railway data become available. The platform retrieves and validates the required information, forwards it to the relevant component, and receives the analytical result.

Results may be stored, shared with another component, or displayed to an authorized user. For example, demand forecasts and delay predictions can be presented together for service assessment and may inform scheduling alternatives where suitable interfaces exist.

The actual level of automation must be supported by implementation and integration evidence.

4.4.2 Invalid or Incomplete Data

Missing fields, invalid identifiers, and incorrectly formatted records should be rejected or flagged. The system should communicate the problem rather than silently substituting arbitrary values.

Partial results should be labelled with their limitations; otherwise, the system should indicate that the result is unavailable.

4.4.3 Component or Service Failure

When a component fails, the platform should report the failure rather than present an unverified result as current. Previously generated results may be shown with timestamps and warnings when appropriate.

Unrelated functions should remain available where the architecture permits. This behaviour requires integration testing.

4.4.4 Conflicting or Outdated Information

Conflicting identifiers, schedules, or operational statuses should be resolved using agreed rules or flagged for review. Predictions should include timestamps or relevant metadata to indicate their freshness.

4.4.5 Fraud Alerts and Operational Safety

Fraud alerts should prioritize investigation rather than automatically establish wrongdoing. Scheduling alternatives must be checked against operational and safety constraints before authorization. When essential information is unavailable, the system should communicate the limitation instead of presenting unsupported recommendations.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Implementation Overview

CeylonRail Intelligence integrates four railway intelligence components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, AI-based fraud detection and passenger verification, and multi-objective train scheduling optimization.

The implementation follows a modular approach, allowing each component to perform its own data preparation, analytical processing, and output generation. Shared platform services support communication and result presentation. The implementation covers dataset preparation, model development, application integration, interface development, and system evaluation.

The implementation is considered at two levels: **component-level implementation**, focusing on individual research functions, and **platform-level integration**, focusing on shared services and communication. The system aims to support railway planning and decision-making rather than replace established operational procedures.

5.2 Development Environment and Technology Stack

5.2.1 Technology Selection

The integrated research design identifies the following technologies.

Table 8 - Technologies and Their Roles

Technology	Main purpose
React.js	Interactive user interface
Tailwind CSS	Interface styling
Leaflet Maps	Geographical visualization
Node.js and Express.js	Backend services and APIs
Python	Data processing and analytical operations
TensorFlow and Scikit-Learn	Machine-learning development and evaluation

NSGA-II	Multi-objective scheduling optimization
MongoDB	Proposed centralized data storage

The individual components use different technologies. For example, the fraud-detection component describes React, Flask, Python, and MySQL, while demand forecasting uses Python libraries and CSV/JSON files. Therefore, the final report must distinguish the planned architecture from the technologies actually implemented. Integration requires compatible interfaces or adapters where necessary.

5.2.2 Frontend Implementation

React.js provides a common interface for accessing forecasts, train status, fraud-risk assessments, and scheduling results. Tailwind CSS supports consistent and responsive layouts, while Leaflet Maps enables geographical visualization when location data are available.

The interface should distinguish observed information from predictions and proposed schedules. It should also display appropriate error messages and restrict sensitive passenger information to authorized users.

5.2.3 Backend and Analytical Processing

Node.js and Express.js are identified for application services and API communication in the integrated design. These services handle requests, validate inputs, retrieve data, and return component results.

Python supports machine-learning and optimization operations, including prediction generation, feature processing, and model evaluation. Where analytical services operate separately, defined interfaces are required to exchange structured requests, results, and processing statuses.

5.2.4 Database and Data Storage

The integrated design proposes MongoDB for centralized railway information, including train details, timetables, historical demand, transactions, and analytical outputs. However, the fraud-detection component describes MySQL, while demand forecasting also uses CSV and JSON files.

These technologies should not be presented as one unified database implementation unless integration or migration has been completed. Multiple storage systems require clear data ownership, consistent identifiers, controlled access, and appropriate backup procedures.

5.3 Implementation of the Four Research Components

5.3.1 Passenger Demand Forecasting and Adaptive Seat Allocation

The demand-forecasting component uses historical railway passenger and ticketing data. Its individual research describes 55,302 records covering 2020–2024. Data preparation includes preprocessing, feature engineering, and preparing records for model development.

The research evaluates Random Forest, XGBoost, and LSTM approaches. Random Forest is identified as the selected forecasting model, while LSTM is used for experimentation rather than the current live prediction service.

Python libraries support data processing, model development, evaluation, and model persistence. Forecasted demand is then used by a PuLP-based optimization process to allocate seats across passenger classes subject to capacity and allocation constraints.

The outputs include demand forecasts, class-wise seat allocation, and utilization information. These results can support scheduling decisions when the relevant data exchange is implemented. Forecasting and allocation should be evaluated separately because accurate predictions do not automatically guarantee effective allocation.

5.3.2 Real-Time Train Tracking and Delay Prediction

This component estimates train delays and arrival times using available operational information, historical records, and location data. The integrated research identifies Random Forest, Gradient Boosting, and LSTM among the evaluated approaches.

The workflow involves preparing input data, applying the selected model, generating predictions, and presenting results through dashboard summaries and maps where suitable location information exists.

Prediction reliability depends on data quality and freshness. Historical or simulated GPS data alone do not establish real-time operational validation. Live deployment requires further testing of location updates, prediction accuracy, and service reliability.

Delay predictions can support operational monitoring and scheduling optimization when the necessary interfaces are implemented.

5.3.3 AI-Based Railway Fraud Detection and Passenger Verification

The fraud-detection component identifies suspicious ticketing activities and supports ticket verification. Its individual research describes approximately 9,800 transaction records with 18 attributes covering 2020–2024 and evaluates Isolation Forest, One-Class SVM, and Autoencoder approaches.

The implementation involves preparing transaction features, generating anomaly scores, classifying risk levels, and applying relevant ticket-validation rules. Outputs may include validation results and low-, medium-, or high-risk classifications.

The individual component describes React, Flask, Python, and MySQL, which differ from the integrated design's proposed Node.js/Express and MongoDB stack. Compatible interfaces are therefore required for integration.

Anomaly scores indicate unusual behaviour rather than confirmed fraud. Results should support authorized staff in prioritizing reviews, with final decisions subject to appropriate verification procedures.

The integrated paper also describes Random Forest and rule-based fraud detection. The final report must clarify which approach is implemented and distinguish it from models evaluated only during individual research.

5.3.4 Multi-Objective Train Scheduling Optimization

The scheduling component evaluates alternative train schedules against operational objectives and constraints. Required inputs include timetable information, routes, train capacities, and the defined optimization criteria.

The integrated design identifies NSGA-II for exploring alternative schedules and examining trade-offs involving delay, congestion, capacity utilization, and resource use, where these objectives are implemented.

The process defines constraints, generates candidate schedules, evaluates their objective values, and compares feasible alternatives. Demand forecasts and delay predictions may provide additional inputs when the corresponding connections exist.

The results support timetable planning rather than automatic operational changes. Any proposed schedule must comply with railway safety requirements and receive appropriate authorization.

5.4 Integrated Application and Component Communication

5.4.1 Application Integration

The integrated application provides a common interface for accessing the four research components while preserving their specialized analytical functions. The frontend sends requests to backend services, which validate inputs, retrieve relevant data, invoke the appropriate component, and return results.

This modular architecture reduces direct dependencies between components and supports independent development and testing.

5.4.2 Cross-Component Data Exchange

Table 9 - Key Cross-Component Data Exchanges

Source	Receiving component	Information exchanged
Demand forecasting	Scheduling optimization	Forecast demand and seat allocation
Delay prediction	Scheduling optimization	Estimated delays and arrival times
Shared railway information	Forecasting and scheduling	Routes, stations, timetables, and capacity
Shared railway information	Fraud detection	Relevant ticket, train, and route information
All components	Common dashboard	Predictions, risk assessments, allocations, and schedules

Consistent train, route, and station identifiers are essential for matching records. Data formats and units should also be standardized or mapped between services.

5.4.3 Handling Integration Errors

The application should handle missing data, invalid identifiers, unavailable services, and incomplete results. Clear error messages should be displayed instead of presenting unsupported information as current.

Older results should include timestamps where applicable, and error messages should not expose sensitive information. These behaviours must be verified through component and integration testing.

5.5 Project Execution and Coordination

5.5.1 Work Organization

The project combines four individual research components with a shared integration objective. Activities include data preparation, model development, evaluation, interface development, integration, and documentation.

Shared responsibilities include defining the architecture, agreeing on data formats, identifying dependencies, checking interface compatibility, and consolidating research findings. Individual model performance alone does not demonstrate successful platform integration.

5.5.2 Coordination and Version Control

Project coordination requires documented decisions, communication of interface changes, and tracking of implementation progress. Version control supports change tracking, code review, and the management of component-specific development.

Datasets, configuration files, and model artefacts should be managed carefully, particularly when they contain sensitive information. The final report should describe only the coordination tools and practices actually used.

5.5.3 Implementation Challenges and Responses

Key challenges include differences in technology stacks, inconsistent data structures, and varying availability of historical, simulated, and live data.

These challenges can be addressed through defined interfaces, shared identifiers, documented assumptions, and systematic testing. The group should also verify model names, dataset sizes, technology choices, and evaluation metrics against the individual theses and integrated research paper.

5.6 Deployment Considerations and Implementation Limitations

5.6.1 Deployment Considerations

Deployment requires an environment capable of hosting the frontend, backend services, analytical components, and data storage. Model files and preprocessing

artefacts must be available to the relevant services, while credentials should be stored securely.

Before deployment, the system should be checked for dependency, configuration, database, and interface errors. Access controls, backups, logging, and recovery procedures should also be considered.

Operational railway deployment requires further validation using real operational data, security and performance testing, user acceptance testing, and confirmation that recommendations comply with railway procedures.

5.6.2 Implementation Limitations

System performance depends on the quality and availability of railway data. Historical records may not fully represent future demand or changing operating conditions, while experimental results may not generalize to live environments.

Differences in component technologies and model approaches also require verification of the final architecture and integration status. Delay prediction needs suitable operational validation, while fraud detection requires careful assessment of false positives and human review.

Therefore, the platform should be considered a foundation for railway decision support rather than evidence that all operational challenges have been resolved.

.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

The evaluation of CeylonRail Intelligence examines the performance of its four research components and the effectiveness of their integration. The framework covers model performance, functional verification, system integration, and operational suitability.

Table 10 - Integrated Evaluation Framework

Evaluation area	Key criteria
Demand forecasting	MAE, RMSE, MAPE
Train tracking and delay prediction	Prediction errors, classification performance, data freshness
Fraud detection	Precision, recall, F1-score, ROC-AUC, false-positive rate
Scheduling optimization	Delay, capacity, congestion, resource utilization
Integrated application	Integration success, response time, availability
Operational validation	Functionality, security, usability, stakeholder feedback

The integrated research paper reports a 99.1% integration success rate, 1.7-second average API response time, 1.4-second dashboard load time, support for over 500 concurrent users, and 99.3% system availability. These are reported results; complete test configurations and measurement procedures are not established by the available summary. They should not be treated as independently verified production guarantees.

6.2 Integration and End-to-End Testing

6.2.1 Testing Approach

Integration testing examines whether the four components communicate correctly through the intended interfaces. Testing should cover functional behaviour, API communication, end-to-end workflows, regression, performance, security, and usability.

The intended workflow begins with data retrieval or user input, followed by component processing and presentation of the results through the dashboard. Cross-component communication requires consistent train, route, station, and transaction identifiers.

The existence of a proposed testing approach does not confirm that every test was executed. Results must be supported by actual test records.

6.2.2 Component and Interface Verification

Table 11 - Integrated Test Matrix

Test ID	Test area	Expected outcome	Evidence status
IT-01	Demand forecasting	Valid inputs produce forecasts	Workflow and model evaluation documented
IT-02	Seat allocation	Capacity and allocation constraints are respected	Constraint-based process described
IT-03	Train tracking	Correct train status and timestamp are presented	Tracking described; live reliability unverified
IT-04	Delay prediction	Valid inputs produce predictions	Experiments documented; classification weakness identified
IT-05	Fraud detection	Risk results are generated	Anomaly-detection experiments documented
IT-06	Ticket verification	Relevant validation checks are applied	Full live validation unverified
IT-07	Scheduling	Candidate schedules satisfy configured constraints	Experimental comparison reported
IT-08	Component integration	Data are exchanged consistently	Complete execution evidence required
IT-09	Performance	Response and loading times are measured	System-level figures reported

IT-10	Error handling	Failures do not produce misleading results	Further evidence required
IT-11	Security	Sensitive information is protected	Formal test results not established
IT-12	User acceptance	Intended users can interpret results	Formal evaluation not established

The matrix summarizes the available evidence and does not imply that every test passed. Test logs, screenshots, API results, and test reports should be used to confirm actual execution status.

6.2.3 End-to-End Behaviour

The intended end-to-end process connects user requests with analytical services and presents the resulting predictions, verification outcomes, or scheduling alternatives.

Integration risks include incompatible data formats, missing records, inconsistent identifiers, and unavailable services. Demand forecasts and delay predictions can support scheduling only when the relevant interfaces are implemented and the exchanged information is correctly associated with each service.

The available documents describe component workflows and overall performance figures but do not establish a complete, independently verifiable end-to-end test report for every cross-component interaction.

6.3 Overall Results

6.3.1 Passenger Demand Forecasting

The individual demand-forecasting thesis evaluates Random Forest and XGBoost using 55,302 historical records covering 2020–2024. Data from 2020–2023 was used for training, while 2024 was reserved for testing.

Table 12 - Demand Forecasting Results

Model	MAE	RMSE	MAPE
Random Forest	24.10	37.67	13.12%
XGBoost	24.20	38.13	13.43%

Random Forest achieved lower errors across all three metrics and was selected as the production forecasting model. The individual thesis also describes an LSTM experiment that is not used by the current live prediction service.

The integrated paper separately reports LSTM results of 42.7 MAE, 54.9 RMSE, and 7.2% MAPE. These results should remain separate because the experimental configurations may differ. A lower MAPE does not necessarily indicate lower absolute error.

The component also uses PuLP for adaptive seat allocation. Its effectiveness requires separate evaluation of constraint satisfaction and comparison against an appropriate allocation baseline.

6.3.2 Train Tracking and Delay Prediction

The integrated research paper reports the following delay-prediction results.

Table 13 - Delay Prediction Results

Model	Accuracy	RMSE (minutes)	R^2
Random Forest	88.1%	5.7	0.86
Gradient Boosting	90.2%	4.9	0.89
LSTM	93.4%	4.1	0.93

LSTM achieved the strongest reported results in the integrated paper. However, the evaluation used simulated GPS information, so these findings do not establish performance under continuous live railway conditions.

The individual thesis reports a separate classification experiment involving 1,065 test observations. All observations were classified as Moderate Delay, resulting in approximately 34.5% accuracy and a weighted F1-score of 0.177.

These results represent different experiments and should not be combined. The classification limitation must be addressed if that model is retained. Further live-data validation is also required before operational use.

6.3.3 Fraud Detection and Passenger Verification

The integrated research paper reports the following results for its Random Forest and rule-based fraud-detection framework.

Table 14 - Integrated Fraud Detection Results

Metric	Reported result
Accuracy	94.6%
Precision	92.8%
Recall	91.5%
F1-score	92.1%
False-positive rate	4.2%

These results suggest promising performance in the reported evaluation. Precision, recall, and false-positive rate are important for balancing the detection of suspicious transactions against unnecessary alerts.

The individual thesis separately evaluates Isolation Forest, One-Class SVM, and Autoencoder models using approximately 9,800 transaction records with 18 attributes. It reports ROC-AUC values of approximately 0.81 for Isolation Forest and 0.96 for both One-Class SVM and Autoencoder. Its 2024 test set contains 2,750 transactions, classified into 2,271 low-risk, 328 medium-risk, and 151 high-risk records.

These results must be reported separately because the models and evaluation configurations differ. An anomalous transaction is not necessarily fraudulent; risk results should support further investigation rather than automatic punitive decisions.

6.3.4 Multi-Objective Train Scheduling Optimization

The integrated paper compares a baseline timetable with an optimized schedule generated using NSGA-II.

Table 15 - Scheduling Optimization Results

Metric	Baseline	Optimized
Average delay	12.1 minutes	6.3 minutes
Capacity utilization	74.2%	88.7%
Route congestion	66.8%	49.5%
Resource utilization	76.3%	90.8%

The reported results indicate:

- Average delay decreased by approximately 48%.
- Capacity utilization increased by 14.5 percentage points.
- Route congestion decreased by 17.3 percentage points.
- Resource utilization increased by 14.5 percentage points.

These improvements demonstrate the potential of the optimization approach under the reported experimental conditions. However, results depend on the input data, baseline schedule, constraints, and objective functions. Further validation across railway routes and operating scenarios is necessary.

6.3.5 Integrated System Performance

Table 16 - Integrated System Performance

Metric	Result
Integration success rate	99.1%
Average API response time	1.7 seconds
Dashboard load time	1.4 seconds
Concurrent users supported	500+

System availability	99.3%

These figures indicate the reported technical performance of the integrated application. However, the test environment, workload, observation period, and measurement methods are not fully established in the available summary.

Overall, the research reports promising component-level results and system performance figures, but live operational validation and comprehensive integration evidence remain limited.

6.4 Stakeholder and User Validation

Stakeholder validation is necessary to determine whether the system meets the practical needs of railway personnel. Technical metrics alone cannot establish whether forecasts, fraud alerts, and scheduling recommendations are understandable and useful.

Potential evaluators include railway operations personnel, timetable planners, station staff, and ticket inspectors. Evaluation could involve task-based testing, questionnaires, and interviews covering train-status retrieval, forecast interpretation, transaction review, and schedule comparison.

Table 17 - Stakeholder Validation Status

Validation area	Evidence required	Current status
Dashboard usability	Task completion and user feedback	Formal results not established
Demand forecasts	Feedback from capacity planners	Formal results not established
Delay information	Evaluation by operations personnel	Formal results not established
Fraud alerts	Review by ticket inspectors	Formal results not established

Scheduling recommendations	Operational review	Formal results not established
Overall acceptance	Documented acceptance testing	Formal results not established

The available documents do not establish formal evaluation of the complete system by railway stakeholders. Therefore, user satisfaction and operational acceptance cannot be claimed as confirmed outcomes. Future evaluation should document participant roles, methods, findings, and improvements.

6.5 Achievement of Common Objectives

Table 18 - Common Objectives

Objective	Assessment
Demand forecasting and seat allocation	Component implementation and experimental evaluation documented; full integration requires confirmation
Train tracking and delay prediction	Research completed, but classification weaknesses and live-data limitations remain
Fraud detection and verification	Analytical experiments documented; final integrated configuration and live performance require confirmation
Multi-objective scheduling	Experimental improvements reported; operational feasibility remains unverified
Common platform integration	System-level metrics reported; complete interface testing requires further evidence
Integrated system evaluation	Quantitative results available; comprehensive testing and stakeholder evidence remain incomplete
Operational deployment readiness	Not established by the available evidence

The project provides a foundation for combining four railway intelligence functions. However, component-level success does not automatically demonstrate that the complete platform is fully integrated or ready for operational use.

6.6 Integrated Limitations and Threats

The evaluation has several important limitations.

Differences between experiments: Individual theses and the integrated paper report different model configurations and results. These should remain separate until the final model versions, datasets, and evaluation procedures are confirmed.

Data quality: Missing values, inconsistent identifiers, and limited coverage may affect model reliability. Historical data may not represent future railway conditions.

Limited live validation: Simulated GPS data and historical transaction records do not establish long-term operational performance.

Model weaknesses: The delay-classification experiment failed to distinguish delay categories. Fraud alerts may also produce false positives and require human review.

Integration dependencies: Different frameworks and databases may affect data exchange. Actual interfaces, error handling, and shared identifiers require verification.

Incomplete testing evidence: The available documentation does not establish that every integration, regression, security, usability, and performance test was completed successfully.

Generalizability: Scheduling and model improvements depend on the experimental datasets, baselines, and constraints. Further testing across routes and operating scenarios is needed.

Stakeholder acceptance: Formal evaluation by railway personnel has not been established, leaving practical usefulness and operational compatibility unconfirmed.

Addressing these limitations requires reconciling component results, documenting reproducible tests, completing integration verification, evaluating suitable operational data, and obtaining structured user feedback.

6.7 Chapter Summary

This chapter evaluated the four research components and the reported performance of the CeylonRail Intelligence platform. Random Forest performed best in the individual demand-forecasting comparison, LSTM achieved the strongest delay-prediction results in the integrated paper, and the fraud-detection experiments reported promising results. The scheduling comparison also showed improvements across the selected indicators.

However, differences between experiments, a significant delay-classification weakness, limited live-data validation, and incomplete integration and stakeholder evidence restrict the conclusions that can be drawn.

Overall, CeylonRail Intelligence demonstrates the potential of integrating forecasting, train tracking, fraud detection, and scheduling optimization into a common decision-support platform. Further technical testing and operational validation are required before the system can be considered ready for real railway deployment.

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

CeylonRail Intelligence is proposed as an AI-based railway decision-support system combining passenger demand forecasting and seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and multi-objective scheduling optimization.

Integrating these functions may help railway personnel make more informed decisions by connecting demand forecasts, operational information, ticket-risk assessments, and scheduling alternatives. However, financial savings and improvements in passenger satisfaction have not been established by the available research.

7.1.1 Primary Beneficiaries

Table 7.1: Potential Beneficiaries

Table 19 - Potential Beneficiaries

Beneficiary	Expected value
Railway management	Improved access to operational information
Timetable planners	Comparison of scheduling alternatives
Capacity planners	Demand-based seat allocation
Station and operations staff	Access to train status and delay predictions
Ticket inspectors	Prioritization of suspicious transactions
Passengers	Potentially more predictable services
Technical teams	Modular system maintenance and development

These benefits depend on data quality, model accuracy, integration, and user adoption.

7.1.2 Value Created by Integration

A common platform may reduce the need to consult separate tools and manually combine information. Demand forecasts can support capacity planning, delay

predictions can inform scheduling, and fraud-risk assessments can help prioritize ticket verification.

The reported experimental results provide an initial technical basis for this value proposition. However, actual financial savings and network-wide service improvements require operational validation and stakeholder evidence.

7.1.3 Social Value

The system may support better railway planning, capacity utilization, and service reliability. However, inaccurate predictions, false fraud alerts, and unsuitable scheduling recommendations could negatively affect passengers. Responsible adoption requires attention to accuracy, accessibility, privacy, safety, and fairness.

7.2 Market Alternatives and Differentiation

7.2.1 Target Market

The primary prospective customer is Sri Lanka's railway operator, with potential involvement from public transport authorities and railway technology providers.

The system is best positioned initially as a decision-support platform that complements existing railway systems. A controlled pilot could evaluate selected functions before wider adoption. Market demand and customer commitment have not yet been established.

7.2.2 Existing Alternatives

Potential alternatives include manual planning, conventional railway information systems, separate tracking tools, rule-based ticket verification, and standalone forecasting or optimization software.

Table 20 - Alternative Approaches and Potential Differentiation

Alternative	Potential distinction

Manual planning and spreadsheets	Adds predictive analysis and optimization
Conventional railway systems	Provides complementary intelligence functions
Standalone tracking tools	Connects tracking with delay prediction
Rule-based ticket verification	Adds anomaly detection and risk scoring
Demand-forecasting tools	Links forecasts with seat allocation
Standalone optimization tools	Combines scheduling with other railway intelligence functions

These distinctions describe the intended design rather than proven competitive advantages. Existing systems may already provide similar capabilities, so further market research is necessary.

7.2.3 Differentiation of the Integrated Solution

The proposed differentiation is the combination of four railway intelligence functions within one modular platform, with an emphasis on Sri Lankan railway requirements.

The modular architecture may support independent component development and staged adoption. However, maintaining different technologies and analytical services may increase integration and maintenance costs.

Further comparison with existing products is required to establish the solution's uniqueness and measurable advantages.

7.3 Business Model and Feasibility

7.3.1 Proposed Business Model

A business-to-government or institutional software model is a potential commercialization pathway. Adoption could begin with a pilot, followed by wider implementation if technical, operational, and financial evaluations are satisfactory.

Table 21 - Business Model Elements

Element	Proposed approach
Customers	Railway operator and public transport authorities
Value proposition	Integrated predictive and optimization support
Delivery	Institutional procurement or approved technology partnership
Key activities	Development, integration, monitoring, and maintenance
Key resources	Software engineers, data scientists, and railway expertise
Partners	Railway authorities, technology providers, and research institutions
Costs	Development, hosting, security, training, and support
Potential funding	Research grants, institutional funding, or software contracts

This is a proposed model; no confirmed customers, revenue, partnerships, or funding arrangements are established.

7.3.2 Technical Feasibility

The four components demonstrate research-level development and experimental evaluation. Nevertheless, differences in model approaches, backend frameworks, and databases require further integration work.

Technical feasibility should be confirmed through interface testing, realistic workload evaluation, external data integration, and failure-recovery testing.

7.3.3 Operational and Financial Feasibility

Operational feasibility depends on access to authorized railway data, suitable infrastructure, staff cooperation, and compatibility with existing procedures.

Financial feasibility remains unverified because detailed development costs, hosting expenses, licensing costs, and measured savings are unavailable. A future assessment should consider:

- Software development and integration.
- Data access and preparation.
- Hosting, storage, and backups.
- Model maintenance and monitoring.
- Security and privacy assessments.
- Training, support, and operational testing.

Potential benefits include reduced manual analysis, improved capacity planning, and better-informed scheduling. These benefits should be measured against an established baseline before financial savings are claimed.

7.4 Adoption, Intellectual Property and Compliance

7.4.1 Adoption Barriers

Adoption may be affected by limited data access, incompatible systems, infrastructure costs, procurement requirements, and organizational readiness.

Railway personnel would require training to interpret predictions and risk scores correctly. The system should complement existing procedures and preserve clear responsibility for operational decisions.

7.4.2 Intellectual Property and Ownership

The project includes software, research documents, model implementations, and interface designs. Ownership and permitted use must be clarified before commercialization.

The individual demand-forecasting thesis grants SLIIT a non-exclusive right to reproduce and distribute that thesis. This statement does not automatically determine ownership of all software, datasets, model artefacts, or group contributions.

The team should review university requirements, project agreements, software licences, and data-provider terms. Commercial use should proceed only after ownership and licensing permissions have been clarified.

7.4.3 Data Protection, Privacy and Security

The platform may process passenger, ticketing, and operational information. Data minimization, anonymization or pseudonymization where appropriate, and restricted access should be applied.

Recommended safeguards include authentication, role-based access control, secure communication, protected credentials, input validation, logging, backups, and defined data-retention periods.

Applicable Sri Lankan data protection requirements, including the Personal Data Protection Act, No. 9 of 2022 and relevant amendments or regulations in force, should be reviewed with qualified guidance before deployment. This chapter does not establish legal compliance.

Sensitive railway operational information should also be protected against unauthorized disclosure.

7.4.4 Responsible AI and Professional Responsibility

Model outputs should be treated as decision-support information rather than guaranteed facts. Forecasts may be inaccurate, delay predictions may become outdated, and fraud detection may produce false positives.

The interface should distinguish observed, scheduled, and predicted information and indicate data freshness where relevant. Human review should remain essential for fraud-related decisions and scheduling recommendations.

Potential bias should also be examined, particularly where data underrepresent certain routes, ticket types, or operating conditions. The project team should document model limitations, preserve reproducible results, protect data, and follow applicable research ethics and professional standards.

7.5 Sustainable Development Goal Contribution

CeylonRail Intelligence may contribute to three United Nations Sustainable Development Goals through improved railway planning and resource management. These contributions remain potential outcomes rather than demonstrated societal impacts.

7.5.1 SDG 9: Industry, Innovation and Infrastructure

The system supports the concept of data-driven infrastructure management through forecasting, tracking, and scheduling optimization. These capabilities may improve operational planning, but network-wide infrastructure improvements have not been demonstrated.

7.5.2 SDG 11: Sustainable Cities and Communities

More reliable railway services may improve access to employment, education, and other opportunities. However, reductions in congestion, emissions, or passenger travel time require evidence from operational deployment.

7.5.3 SDG 12: Responsible Consumption and Production

Scheduling optimization may improve the use of available railway capacity and resources.

The integrated research paper reports capacity utilization increasing from 74.2% to 88.7% and resource utilization increasing from 76.3% to 90.8%. These are experimental results and do not independently establish reductions in energy use or environmental impact.

7.5.4 Potential Negative Effects and Sustainability Measures

Table 22 - Sustainability Measures

SDG	Potential contribution	Required measure

SDG 9	Improved infrastructure planning	Validate reliability and maintain system performance
SDG 11	Potentially more reliable public transport	Evaluate service quality and fairness
SDG 12	More effective capacity and resource use	Measure real resource consumption and preserve safety constraints

Potential negative effects include inaccurate predictions, unnecessary fraud alerts, unequal model performance, and increased operational complexity. These risks require monitoring, human oversight, and ongoing evaluation.

7.5.5 Long-Term Sustainability

Long-term sustainability depends on reliable data access, model maintenance, security updates, technical support, and clear organizational responsibility.

A sustainable implementation should establish procedures for monitoring model performance, maintaining data quality, responding to incidents, updating software, and recovering from failures.

Financial sustainability requires comparing ongoing costs with measured operational benefits. Organizational sustainability requires staff training, stakeholder engagement, and compatibility with existing railway procedures.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

Railway transportation is an essential part of Sri Lanka's public transport system. Effective railway management requires coordination between passenger demand, train movement, ticket verification, and timetable planning. This research addresses the need for an integrated framework that combines predictive analytics, risk assessment, and optimization to support data-driven railway operations.

The project proposed **CeylonRail Intelligence**, an AI-driven railway decision-support system comprising four components: passenger demand forecasting and adaptive seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and multi-objective train scheduling optimization.

The demand forecasting component evaluated Random Forest and XGBoost using 55,302 historical records from 2020–2024. Random Forest achieved an MAE of 24.10, RMSE of 37.67, and MAPE of 13.12%, making it the preferred model in the individual thesis. Its forecasts can support seat allocation according to expected demand and capacity constraints.

The train tracking and delay prediction component aims to improve awareness of train movement and service disruptions. The integrated paper reports that the LSTM model achieved 93.4% accuracy, an RMSE of 4.1 minutes, and an R^2 of 0.93. However, a separate classification experiment assigned all test observations to the Moderate Delay category, achieving approximately 34.5% accuracy and a weighted F1-score of 0.177. These results represent different experiments. The classification weakness and use of simulated GPS data highlight the need for further improvement and live-data validation.

The fraud detection component supports the identification of suspicious ticketing transactions. The integrated paper reports 94.6% accuracy, 92.8% precision, 91.5% recall, a 92.1% F1-score, and a 4.2% false-positive rate. Separately, the individual thesis reports ROC-AUC values of approximately 0.81 for Isolation Forest and 0.96

for both One-Class SVM and Autoencoder. These results must remain associated with their respective experiments. Risk scores should support further investigation rather than be treated as proof of fraud.

The scheduling component evaluates alternative timetables against operational objectives. The integrated paper reports that optimization reduced average delay from 12.1 to 6.3 minutes, increased capacity utilization from 74.2% to 88.7%, reduced route congestion from 66.8% to 49.5%, and increased resource utilization from 76.3% to 90.8%. These results are promising within the reported evaluation but do not guarantee equivalent improvements under live railway conditions.

At the platform level, the integrated paper reports a 99.1% integration success rate, a 1.7-second average API response time, a 1.4-second dashboard load time, support for over 500 concurrent users, and 99.3% availability. However, the available documentation does not establish sufficient testing details to treat these figures as independently verified production guarantees.

Overall, the project demonstrates the potential of combining four railway intelligence functions within a common decision-support framework. However, differences in model results, technology stacks, and validation levels remain. Therefore, the research problem has been **addressed at the research and prototype level**, but real-world effectiveness has not yet been conclusively established. Further implementation, operational testing, and stakeholder validation are necessary before wider deployment.

8.2 Principal Integrated Contribution

The principal contribution of CeylonRail Intelligence is a modular framework that brings four complementary railway functions into a common decision-support concept.

First, passenger demand forecasting is linked to adaptive seat allocation, connecting predictive analytics with capacity planning. Second, train tracking and delay prediction aim to improve awareness of service conditions and support operational decisions. Third, fraud-risk assessment and passenger verification provide a basis for identifying

transactions requiring further review, with appropriate human oversight and privacy protection. Fourth, multi-objective scheduling optimization evaluates trade-offs involving delay, capacity, congestion, and resource utilization.

The project also establishes an architectural perspective for coordinating component services, data requirements, and user-facing interfaces. Its findings highlight that successful integration requires verified data exchange, consistent interfaces, and traceable model evaluations—not merely a common design.

Although the research provides a foundation for railway decision support in Sri Lanka, it does not yet demonstrate national-level improvements in reliability, passenger satisfaction, financial savings, or environmental performance. These outcomes require further controlled evaluation.

8.3 Future Directions

Further development should prioritize the following areas.

8.3.1 Reconcile Implementations and Evaluation Results

Review the implemented code, datasets, model artefacts, and test records to establish a consistent account of the final system. Document component-specific technologies, interfaces, data mappings, model versions, and evaluation methods to improve reproducibility and resolve inconsistencies between the individual theses and integrated paper.

8.3.2 Improve Data Quality and Use Operational Data

Seek authorized access to suitable railway data, including train locations, timetables, passenger demand, and ticketing records. Check for missing values, duplicates, inconsistent identifiers, timestamp errors, and changing operating patterns. Where approval is obtained, conduct controlled evaluations using live or near-live data.

8.3.3 Strengthen Demand Forecasting and Seat Allocation

Evaluate forecasting performance across routes, passenger classes, peak periods, holidays, and seasonal changes. Compare models using consistent datasets and metrics. Separately assess seat allocation against defined baselines, capacity constraints, utilization, and relevant fairness requirements.

8.3.4 Improve Delay Prediction and Tracking

Investigate the classification component's weak performance by examining class imbalance, feature quality, label definitions, and model suitability. Report per-class precision, recall, F1-scores, confusion matrices, and baseline comparisons. Assess tracking accuracy, update frequency, missing observations, and stale information, ensuring predictions display clear timestamps and data-quality warnings.

8.3.5 Strengthen Fraud Detection and Responsible Verification

Compare fraud-detection models using consistent data and evaluation procedures. Assess precision, recall, F1-score, ROC-AUC, false-positive rates, and the workload generated by alerts rather than relying on accuracy alone. Protect personal information, review performance across relevant transaction types, and ensure risk scores support human investigation rather than automatic conclusions.

8.3.6 Expand Scheduling Evaluation

Test optimized schedules across different routes, demand levels, timetable scenarios, and disruptions. Ensure that optimization respects railway safety rules and operational constraints. Use repeatable baseline comparisons and report schedule feasibility, objective trade-offs, and sensitivity to uncertain inputs.

8.3.7 Complete Integration and Stakeholder Testing

Conduct documented tests of component interfaces, end-to-end workflows, error handling, security, performance, and recovery. Record expected and actual outcomes for each test. Performance results should specify the test environment, workload, duration, and measurement method.

Subject to approval, evaluate the system with railway personnel through task-based testing, interviews, or questionnaires. Use feedback to improve the interface, prediction interpretation, fraud-review process, and scheduling workflow.

8.3.8 Assess Deployment and Long-Term Sustainability

Before deployment, assess infrastructure, data access, security, training, maintenance, support, and recurring costs. A controlled pilot should compare operational outcomes against a defined baseline using indicators such as forecast error, delay-prediction performance, schedule reliability, capacity utilization, false-positive workload, user acceptance, and system availability.

Clear responsibilities should also be established for software maintenance, model monitoring, data governance, security updates, and incident response. Environmental and societal benefits should only be reported when supported by appropriate measurements.

In conclusion, CeylonRail Intelligence provides a foundation for integrated, AI-driven railway decision support in Sri Lanka. Reconciling implementation differences, improving model reliability, validating component interactions, and conducting controlled evaluations with operational data and users are essential steps towards a dependable and responsible solution.

REFERENCES

[1] K. Jitesh, G. Lakhina, L. S. Kolhar, A. Harshitha, and K. Taneja, "Track Wise: Smart Railway Management," in *Proc. 2026 International Conference on AI-Driven Innovations & Applications (IC-AIDA)*, 2026, doi: 10.1109/IC-AIDA68291.2026.11564491.

[2] K. Silva *et al*., "SmartRail–Optimized Train Management System for Sri Lanka Railways," in *Proc. 2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, 2024.

[3] R. Tang *et al*., "A literature review of artificial intelligence applications in railway systems," *Transportation Research Part C: Emerging Technologies*, vol. 140, Art. no. 103679, 2022, doi: 10.1016/j.trc.2022.103679.

[4] S. Sarp, M. Kuzlu, V. Jovanovic, Z. Polat, and O. Guler, "Digitalization of railway transportation through AI-powered services: Digital twin trains," *European Transport Research Review*, vol. 16, Art. no. 58, 2024, doi: 10.1186/s12544-024-00679-5.

[5] Y. Ou, A.-S. Mihăiţă, A. Ellison, T. Mao, S. Lee, and F. Chen, "Rail digital twin and deep learning for passenger flow prediction using mobile data," *Electronics*, vol. 14, no. 12, Art. no. 2359, 2025, doi: 10.3390/electronics14122359.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[7] X. Ma, Z. Tao, Y. Wang, H. Yu, and Y. Wang, "Long short-term memory neural network for traffic speed prediction using remote microwave sensor data," *Transportation Research Part C: Emerging Technologies*, vol. 54, pp. 187–197, 2015, doi: 10.1016/j.trc.2015.03.014.

[8] B. Yu, H. Yin, and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *Proc. 27th International Joint Conference on Artificial Intelligence (IJCAI)*, 2018, pp. 3634–3640, doi: 10.24963/IJCAI.2018/505.

[9] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *Proc. International Conference on Learning Representations (ICLR)*, 2018.

[10] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.

[11] P. Lapamonpinyo, S. Derrible, and F. Corman, "Real-time passenger train delay prediction using machine learning: A case study with Amtrak passenger train

routes," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 3, pp. 539–550, 2022, doi: 10.1109/OJITS.2022.3194879.

[12] D. Zhang, Y. Peng, Y. Zhang, D. Wu, H. Wang, and H. Zhang, "Train time delay prediction for high-speed train dispatching based on spatio-temporal graph convolutional network," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 3, pp. 2434–2444, 2022, doi: 10.1109/TITS.2021.3097064.

[13] M. F. Rigan, A. H. Nur, R. B. Sulaiman, U. Butt, and S. Zaman, "Blockchain-based railway ticketing: Enhancing security and efficiency with QR codes and smart contract," in *Proc. 2025 4th International Conference on Computing and Information Technology (ICCIT)*, 2025, doi: 10.1109/ICCIT63348.2025.10989409.

[14] P. Singh and M. Singh, "Fraud detection by monitoring customer behavior and activities," *International Journal of Computer Applications*, vol. 111, no. 11, pp. 23–32, 2015, doi: 10.5120/19584-1340.

[15] S. Nawghare, R. K. Somkunwar, and Z. Shaikh, "Indian Railways smart ticketing validation system with improved alert approach," in *Proc. 2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, 2023.

[16] S. Patil, S. Shah, S. Trivedi, J. Jani, and P. Kanani, "Digitized railway ticket verification using facial recognition," in *Proc. 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS)*, 2021, doi: 10.1109/ICICCS51141.2021.9432371.

[17] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[18] E. Setyawan and D. D. Damayanti, "Integrated railway timetable scheduling optimization model and rescheduling recovery optimization model: A systematic literature review," in *Proc. 2018 5th International Conference on Industrial Engineering and Applications (ICIEA)*, 2018, doi: 10.1109/IEA.2018.8387101.

[19] D. E. Goldberg, *Genetic Algorithms in Search, Optimization, and Machine Learning*. Reading, MA, USA: Addison-Wesley, 1989.

[20] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.